

## SWAN RIVER, MB, Canada

WMO: 714430

Lat: 52.1151N

Lon: 101.2330W

Elev: 335

StdP: 97.37

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |            |              | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|------------|--------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%         |              |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS         | MCDB         | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-32.8 | (c)<br>-30.4 | (d)<br>-38.6                  | (e)<br>0.1 | (f)<br>-31.4 | (g)<br>-35.9 | (h)<br>0.1 | (i)<br>-30.2 | (j)<br>9.6            | (k)<br>-13.0 | (l)<br>8.8 | (m)<br>-12.6 | (n)<br>3.0            | (o)<br>220 | (p)<br>0.658 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>12.0            | (c)<br>29.4     | (d)<br>19.9 | (e)<br>27.5 | (f)<br>18.7 | (g)<br>25.9 | (h)<br>17.8 | (i)<br>21.4         | (j)<br>27.2 | (k)<br>20.2 | (l)<br>25.5 | (m)<br>19.0 | (n)<br>24.1 | (o)<br>4.1           | (p)<br>220 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.4                            | 14.8 | 24.6 | 18.2 | 13.6 | 22.9 | 17.1 | 12.7 | 21.8 | 64.2          | 27.5 | 59.5 | 25.5 | 55.6 | 24.2 | 27.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 9.2           | (b) 8.1 | (c) 7.3 | (e) -36.4                  | (f) 32.6 | (g) 2.0            | (h) 1.4 | (i) -37.8                                          | (j) 33.6 | (k) -38.9  | (l) 34.5 | (m) -40.0  | (n) 35.3 | (o) -41.4  | (p) 36.3 |    |    |
| (a) 9.2           | (b) 8.1 | (c) 7.3 | (e) -36.5                  | (f) 23.9 | (g) 2.0            | (h) 1.6 | (i) -38.0                                          | (j) 25.1 | (k) -39.1  | (l) 26.0 | (m) -40.2  | (n) 26.9 | (o) -41.7  | (p) 28.1 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 1.8    | -16.8 | -14.8 | -7.7  | 2.5   | 10.0  | 15.5  | 18.3  | 17.1  | 12.0  | 4.2   | -5.9  | -13.9 |  |  |
|                                                                        | DBStd       | 13.90  | 8.46  | 7.66  | 7.69  | 6.20  | 4.95  | 3.73  | 3.02  | 3.37  | 4.27  | 4.65  | 6.77  | 7.94  |  |  |
|                                                                        | HDD10.0     | 3807   | 829   | 694   | 548   | 233   | 64    | 3     | 0     | 1     | 27    | 188   | 477   | 742   |  |  |
|                                                                        | HDD18.3     | 6118   | 1088  | 927   | 806   | 475   | 262   | 98    | 38    | 64    | 194   | 439   | 727   | 1001  |  |  |
|                                                                        | CDD10.0     | 811    | 0     | 0     | 0     | 7     | 62    | 169   | 259   | 219   | 87    | 8     | 0     | 0     |  |  |
|                                                                        | CDD18.3     | 84     | 0     | 0     | 0     | 0     | 2     | 14    | 38    | 25    | 4     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 1076   | 0     | 0     | 0     | 3     | 68    | 183   | 409   | 322   | 88    | 3     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 250    | 0     | 0     | 0     | 1     | 14    | 42    | 93    | 82    | 19    | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.3    | 3.4   | 3.4   | 3.3   | 3.5   | 3.8   | 3.2   | 2.9   | 2.8   | 3.1   | 3.2   | 3.3   | 3.3   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 542    | 19    | 12    | 19    | 29    | 55    | 90    | 89    | 69    | 50    | 40    | 22    | 20    |  |  |
|                                                                        | PrecMax     | 653    | 42    | 39    | 63    | 66    | 131   | 174   | 214   | 182   | 110   | 107   | 74    | 39    |  |  |
|                                                                        | PrecMin     | 348    | 1     | 0     | 2     | 5     | 7     | 21    | 22    | 22    | 5     | 10    | 4     | 5     |  |  |
|                                                                        | PrecStd     | 81     | 12    | 9     | 14    | 16    | 30    | 40    | 46    | 35    | 27    | 20    | 15    | 10    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 3.6   | 3.3   | 10.1  | 22.8  | 28.2  | 30.6  | 31.4  | 31.7  | 29.1  | 21.8  | 11.4  | 4.2   |  |  |
|                                                                        |             | MCWB   | 1.4   | 1.2   | 5.7   | 11.9  | 15.3  | 19.6  | 21.8  | 21.0  | 17.4  | 13.4  | 6.4   | 1.4   |  |  |
|                                                                        | 2%          | DB     | 1.2   | 1.0   | 6.4   | 18.7  | 25.2  | 27.6  | 29.1  | 29.0  | 25.9  | 17.5  | 7.7   | 1.4   |  |  |
|                                                                        |             | MCWB   | -0.5  | -0.4  | 3.3   | 9.7   | 13.7  | 18.5  | 20.8  | 20.2  | 16.5  | 10.4  | 4.2   | -0.5  |  |  |
|                                                                        | 5%          | DB     | -1.7  | -0.9  | 4.3   | 15.5  | 22.3  | 25.4  | 27.3  | 26.8  | 23.1  | 14.3  | 4.6   | -1.3  |  |  |
|                                                                        |             | MCWB   | -2.9  | -2.1  | 1.9   | 8.3   | 12.4  | 16.8  | 19.5  | 18.7  | 15.1  | 8.7   | 2.1   | -2.6  |  |  |
|                                                                        | 10%         | DB     | -4.4  | -3.4  | 2.4   | 12.3  | 19.9  | 23.4  | 25.7  | 24.8  | 20.2  | 11.5  | 2.5   | -3.3  |  |  |
|                                                                        |             | MCWB   | -5.3  | -4.5  | 0.7   | 6.5   | 11.3  | 15.8  | 18.7  | 17.5  | 13.9  | 7.4   | 0.8   | -4.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 1.6   | 1.4   | 5.7   | 12.6  | 17.0  | 22.2  | 23.6  | 22.8  | 19.3  | 13.9  | 7.1   | 1.6   |  |  |
|                                                                        |             | MCDB   | 3.4   | 2.8   | 9.6   | 21.6  | 25.2  | 27.7  | 29.2  | 29.4  | 26.1  | 19.7  | 10.4  | 3.9   |  |  |
|                                                                        | 2%          | WB     | -0.3  | -0.4  | 3.4   | 10.4  | 14.8  | 19.7  | 21.7  | 21.0  | 17.5  | 11.2  | 4.4   | -0.4  |  |  |
|                                                                        |             | MCDB   | 1.1   | 0.7   | 5.9   | 17.5  | 22.5  | 25.1  | 27.2  | 27.2  | 23.6  | 16.1  | 7.5   | 1.2   |  |  |
|                                                                        | 5%          | WB     | -2.8  | -2.4  | 2.0   | 8.5   | 13.5  | 18.2  | 20.6  | 19.7  | 16.0  | 9.3   | 2.3   | -2.4  |  |  |
|                                                                        |             | MCDB   | -1.6  | -1.3  | 3.8   | 14.6  | 20.5  | 23.1  | 25.9  | 25.1  | 21.5  | 13.3  | 4.3   | -1.3  |  |  |
|                                                                        | 10%         | WB     | -5.2  | -4.7  | 0.7   | 6.9   | 12.1  | 16.9  | 19.4  | 18.6  | 14.4  | 7.9   | 0.9   | -4.2  |  |  |
|                                                                        |             | MCDB   | -4.4  | -3.7  | 2.2   | 12.0  | 18.4  | 21.9  | 24.3  | 23.4  | 19.2  | 11.1  | 2.4   | -3.2  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 8.9   | 9.9   | 10.5  | 11.5  | 12.9  | 11.7  | 12.0  | 12.7  | 12.6  | 9.7   | 7.9   | 8.1   |  |  |
|                                                                        |             | MCDBR  | 9.8   | 9.8   | 10.5  | 17.1  | 17.6  | 15.3  | 14.8  | 16.1  | 17.8  | 15.5  | 10.7  | 9.9   |  |  |
|                                                                        |             | MCWBR  | 8.6   | 8.7   | 8.0   | 9.5   | 8.4   | 7.4   | 7.8   | 8.1   | 9.3   | 9.2   | 7.7   | 8.6   |  |  |
|                                                                        |             | MCDBR  | 9.6   | 9.7   | 9.8   | 15.6  | 15.1  | 13.1  | 13.4  | 14.3  | 16.0  | 14.2  | 10.0  | 9.4   |  |  |
|                                                                        | 5% WB       | MCWBR  | 8.6   | 8.7   | 7.6   | 9.1   | 7.8   | 7.1   | 7.8   | 7.9   | 8.8   | 8.7   | 7.4   | 8.3   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.263 | 0.262 | 0.304 | 0.340 | 0.349 | 0.371 | 0.381 | 0.374 | 0.339 | 0.294 | 0.264 | 0.247 |  |  |
|                                                                        | taud        |        | 2.238 | 2.291 | 2.290 | 2.375 | 2.409 | 2.371 | 2.376 | 2.381 | 2.466 | 2.554 | 2.457 | 2.300 |  |  |
|                                                                        | Ebn at Noon |        | 741   | 868   | 890   | 898   | 903   | 882   | 866   | 852   | 847   | 831   | 761   | 707   |  |  |
|                                                                        | Edh at Noon |        | 65    | 83    | 103   | 108   | 110   | 115   | 113   | 106   | 87    | 65    | 53    | 51    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.15  | 2.31  | 3.80  | 5.00  | 5.57  | 5.63  | 5.97  | 4.87  | 3.52  | 2.00  | 1.02  | 0.81  |  |  |
|                                                                        | RadStd      |        | 0.09  | 0.14  | 0.23  | 0.39  | 0.52  | 0.44  | 0.28  | 0.42  | 0.35  | 0.27  | 0.15  | 0.09  |  |  |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (10 neighbors) | N/A   | N/A     | +2.14  | N/A     | N/A   | -0.55 | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | -0.45 | -0.57 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page